

3.1.2 Sensing

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i> (i) current as the flow of charged particles (ii) potential difference as energy per unit charge (iii) resistance and conductance, including series and parallel combinations (iv) the effect of internal resistance and the meaning of e.m.f. (v) dissipation of power in electric circuits (vi) the relationship between potential difference and current in ohmic resistors (Ohm's law) (vii) the action of a potential divider (viii) simple electrical behaviour of metals, semiconductors and insulators in terms of the number density of mobile charge carriers (ix) conservation of charge and energy.	HSW1, 7 HSW9, 12 HSW9, 10, 12 as represented by Kirchhoff's first and second law HSW1
(b) <i>Make appropriate use of:</i> (i) the terms: e.m.f, potential difference, current, charge, resistance, conductance, series, parallel, internal resistance, load, resistivity, conductivity, charge carrier number density (ii) and recognise standard circuit symbols <i>by sketching and interpreting:</i> (iii) graphs of current against potential difference and graphs of resistance or conductance against temperature for ohmic and non-ohmic devices or components.	M3.12 HSW11
(c) <i>Make calculations and estimates involving:</i> (i) $R = \frac{V}{I}$, $G = \frac{I}{V}$, $V = \frac{W}{Q} = \frac{P}{I}$, $P = IV = I^2 R$, $W = VIt$,	Learners will also be expected to recall the equations for R and G M0.4, M0.1, M0.5, M1.1, M2.3 Epsilon is used as the symbol for e.m.f. to avoid confusion with E which is used for energy and electric field. The ASE guide 'Signs symbols and systematics' details E as the correct symbol for e.m.f. and this will be credited in all examinations

$$(ii) \quad I = \frac{\Delta Q}{\Delta t},$$

$$\frac{1}{G} = \frac{1}{G_1} + \frac{1}{G_2} + \dots \quad G = G_1 + G_2 + \dots$$

$$R = R_1 + R_2 + \dots \quad \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$(iii) \quad R = \frac{\rho L}{A}; G = \frac{\sigma A}{L}$$

(iv) simple cases of a potential divider in a circuit using:

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} \times V_{\text{in}} \quad \text{and} \quad \frac{V_1}{V_2} = \frac{R_1}{R_2}.$$

M1.4, M2.3, M4.3

Learners will be expected to recall the proportionality of potential difference and resistance in a series circuit

(d) *Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):*

(i) investigating electrical characteristics for a range of ohmic and non-ohmic components using voltmeters and ammeters

links to **3.1.2a(vi), b(iii)**
PAG3

(ii) determining the resistivity or conductivity of a metal

links to **3.1.2c(iii)**
PAG3

(iii) use of potential divider circuits, which may include sensors such as thermistor, LDR

links to **3.1.2a(vii)**
PAG4

(iv) the calibration of a sensor or instrument

links to **3.1.2a(vii), c(iii)**
PAG3

(v) determining the internal resistance of a chemical cell or other source of e.m.f.

links to **3.1.2a(iv)**
PAG3